

1. **The Dimension:** The dimension is the numerical value that is being assigned to the size, shape or location of the feature being described.

2. **Dimension Line:** Dimension lines are thin continuous line that show the extent and direction of the dimension. Arrowhead are placed at the ends of dimension line.

3. **Extension Line:** An extension line is also a thin continuous line drawn in extension of an outline. It extends by about 2mm to 4mm (approximately) beyond the dimension line.

4. **Arrow head:** An arrow head is placed at each end of a dimension line. Its pointed touches an outline, an extension line or centre line. The size of an arrowhead should be proportion to the thickness of the outline. The length of the arrowhead should be about three times its maximum width. It is drawn freehand with two strokes made in the direction of its pointed end. The space between them is neatly filled up.

5. **Leader line:** A leader line is a thin continuous line that is used to associate a note, symbol or dimension with a feature. Leader lines are typically used to call out radii and diameters. Leader lines are inclined to the reference line XY usually 30° to 45° or 60° or 75° .

6. **Note:** A note give information regarding specific operation relating to a feature. It is placed outside a view but adjacent to the feature concerned. It is so written that it may be read when the drawing is viewed from the bottom edge.

Systems of Dimensioning:

There are two systems of dimensioning:

1. Aligned System
2. Unidirectional System

1. Aligned System:

In the aligned system the dimension is placed perpendicular to the dimension line in such a way that it may be read from bottom edge or the right hand edge of the drawing sheet. The dimensions should be placed near the middle and above, but clear of the dimension line.

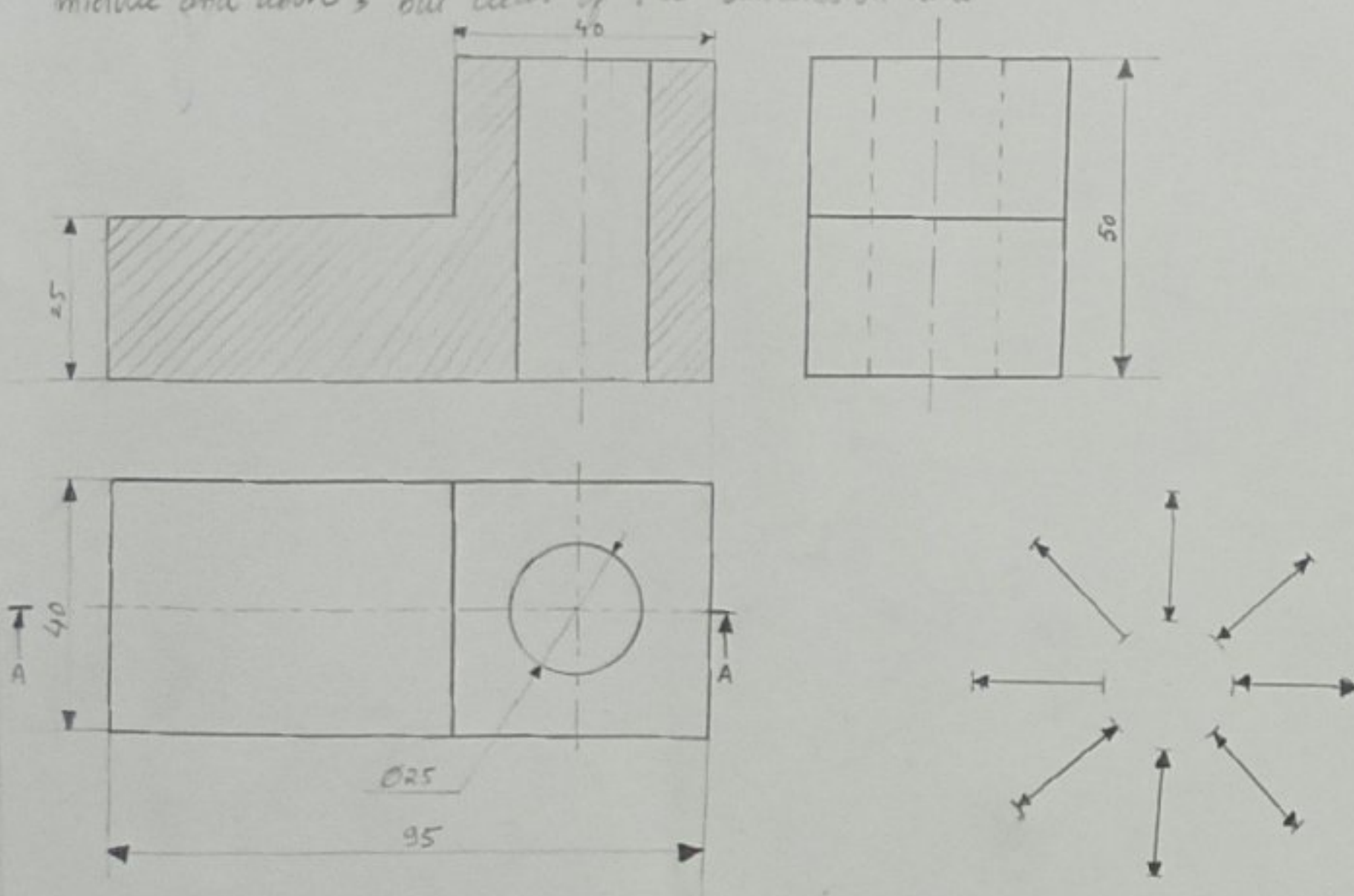


Fig: 2

2. Unidirectional Systems:-

In unidirectional system all dimensions are so placed that they can be read from the bottom edge of the drawing sheet. The dimension lines are broken near the middle for inserting the dimensions.

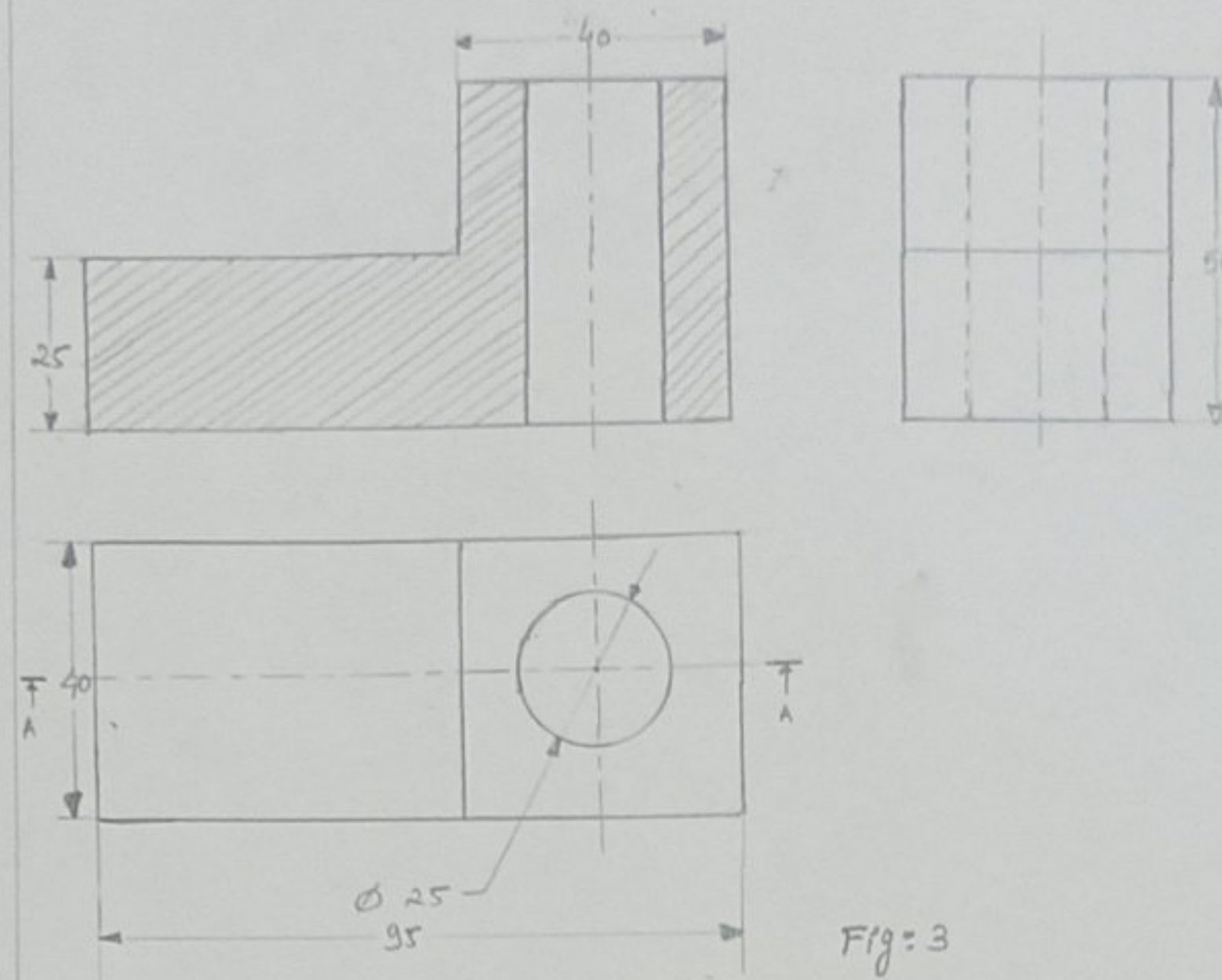


Fig: 3

Out of two dimensioning systems, aligned system of dimensioning is preferred. Aligned system dimensioning may be done in following ways:

1. **Parallel Dimensioning:** parallel dimensioning consists of several dimensions originating from one projection line.

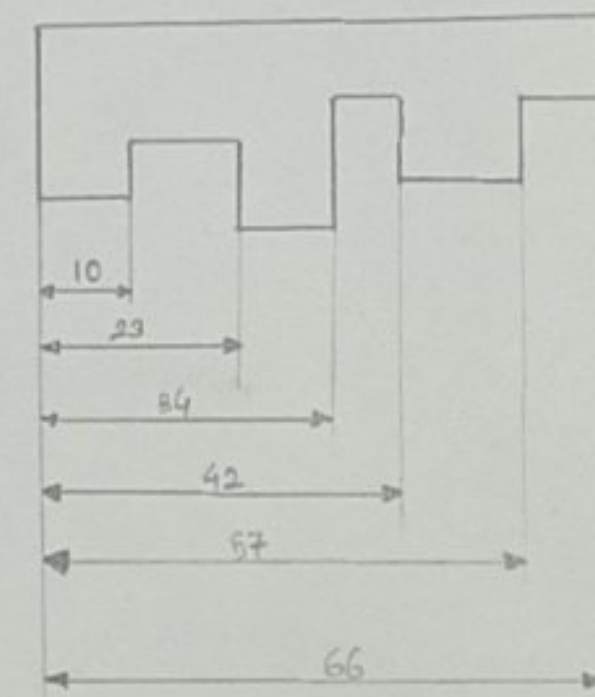


Fig: 4

2. **Chain Dimensioning:** The dimension are arranged in a straight line.

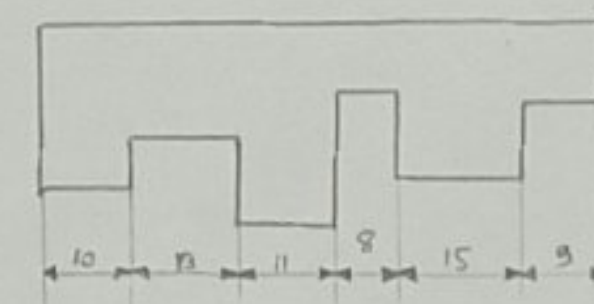


Fig: 5

3. **Combined Dimensioning:** The simultaneous use of chain and parallel dimensioning results in combined dimensioning.

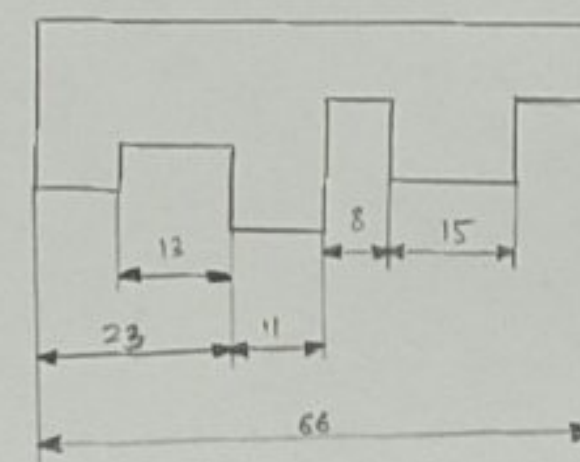


Fig: 6

4. **Co-ordinate Dimensioning:** In co-ordinate dimensioning a baseline or datum line is determined for each cartesian co-ordinate direction and all dimensions are specified with respect to these base lines.

Dimensions may be given with dimension lines and arrow head (Refer fig 7 (a)) or without dimension lines and arrow heads (Refer fig 7(b))

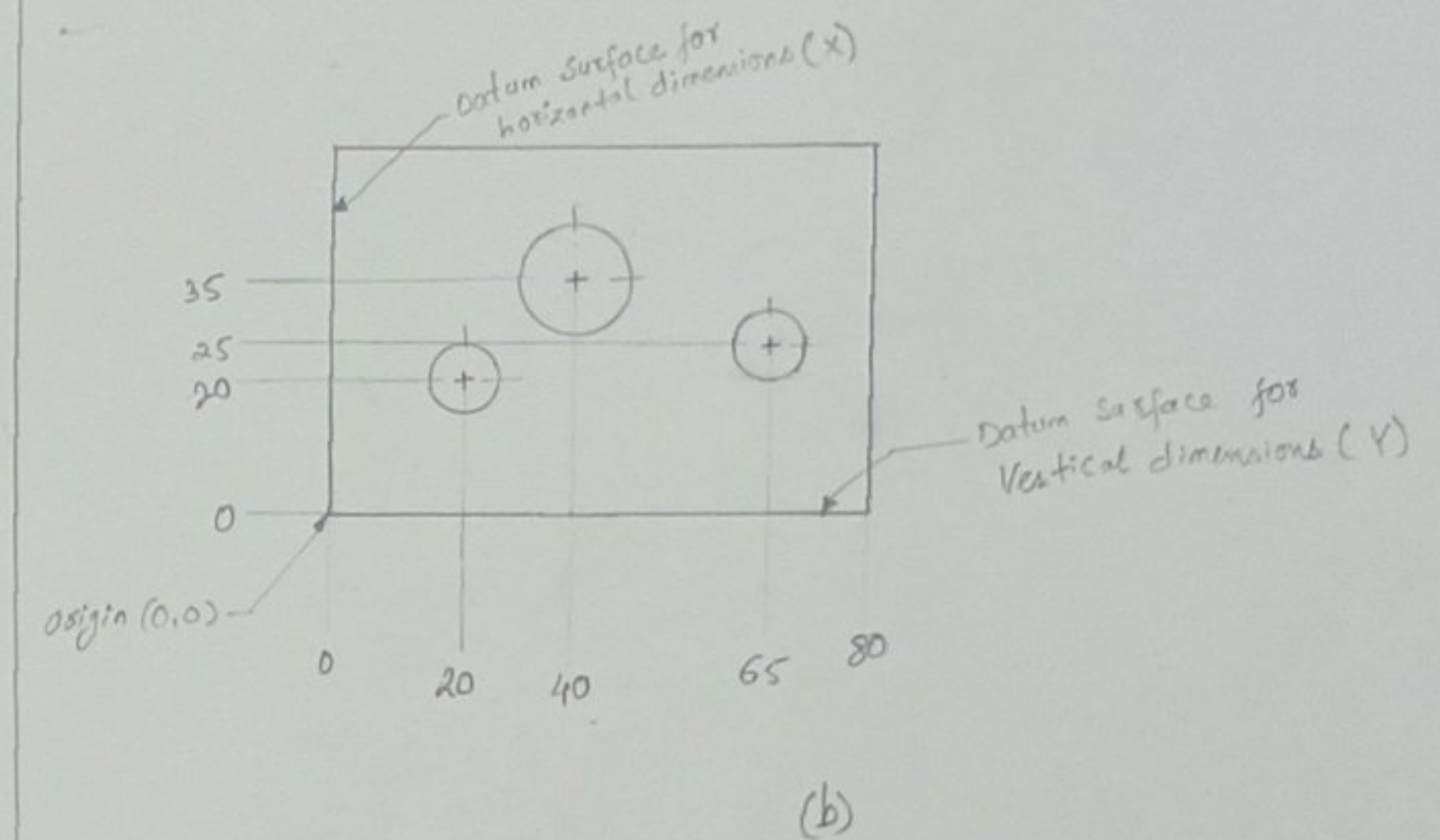
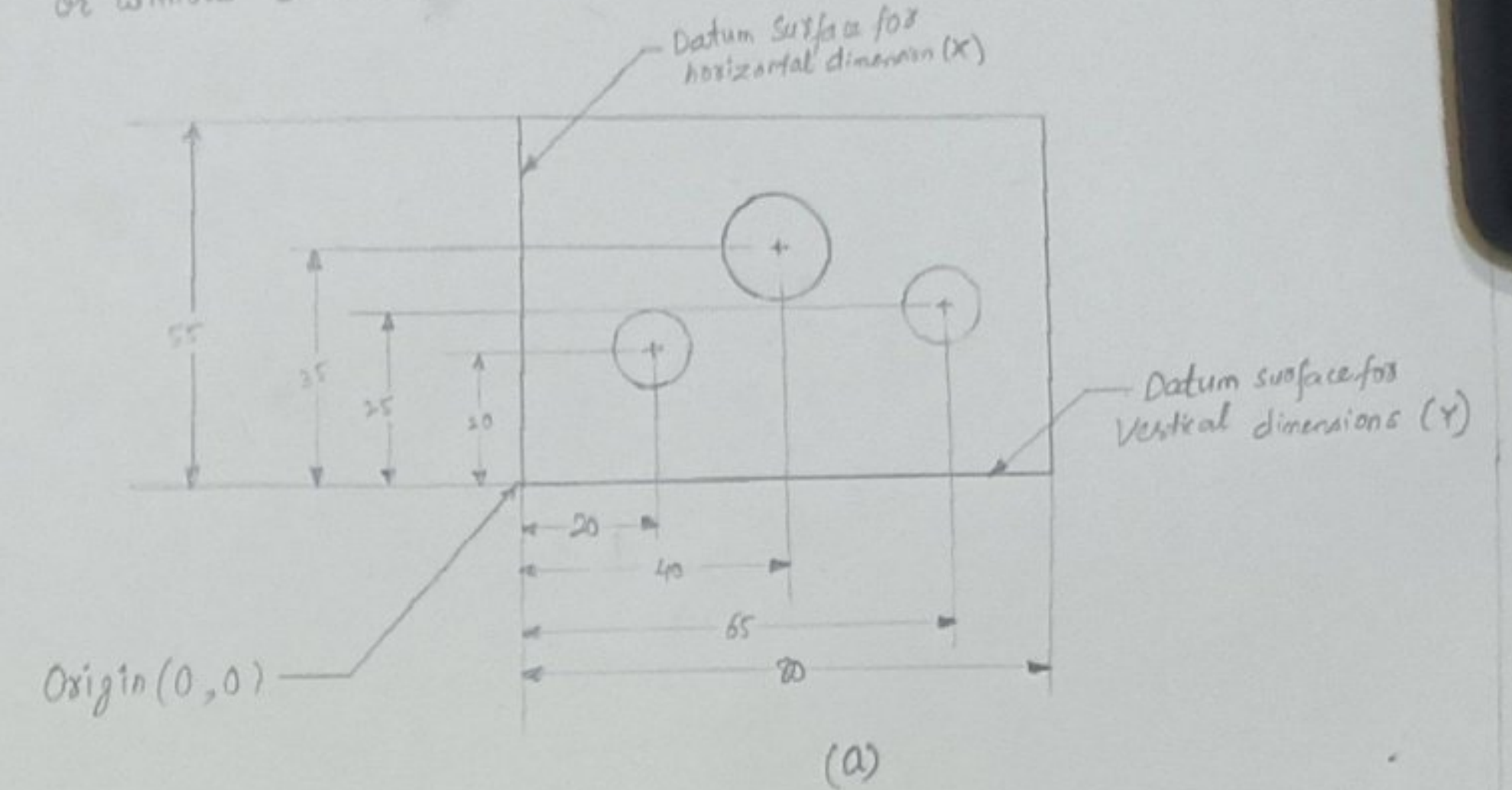


Fig: 7

GRAMIN TECH.3 MANAG. CAMPUS VISNUPURI NED

NABEEL AHMED AHMED MUBASSIR

TITLE: SHEET ON

CLASS: FYBTECH (Comp. Engg) ROLL NO: 24COB67

DATE: SHEET NO: 02 SCALE: -

METHOD OF PROJECTION: SIGN: MARKS: 100

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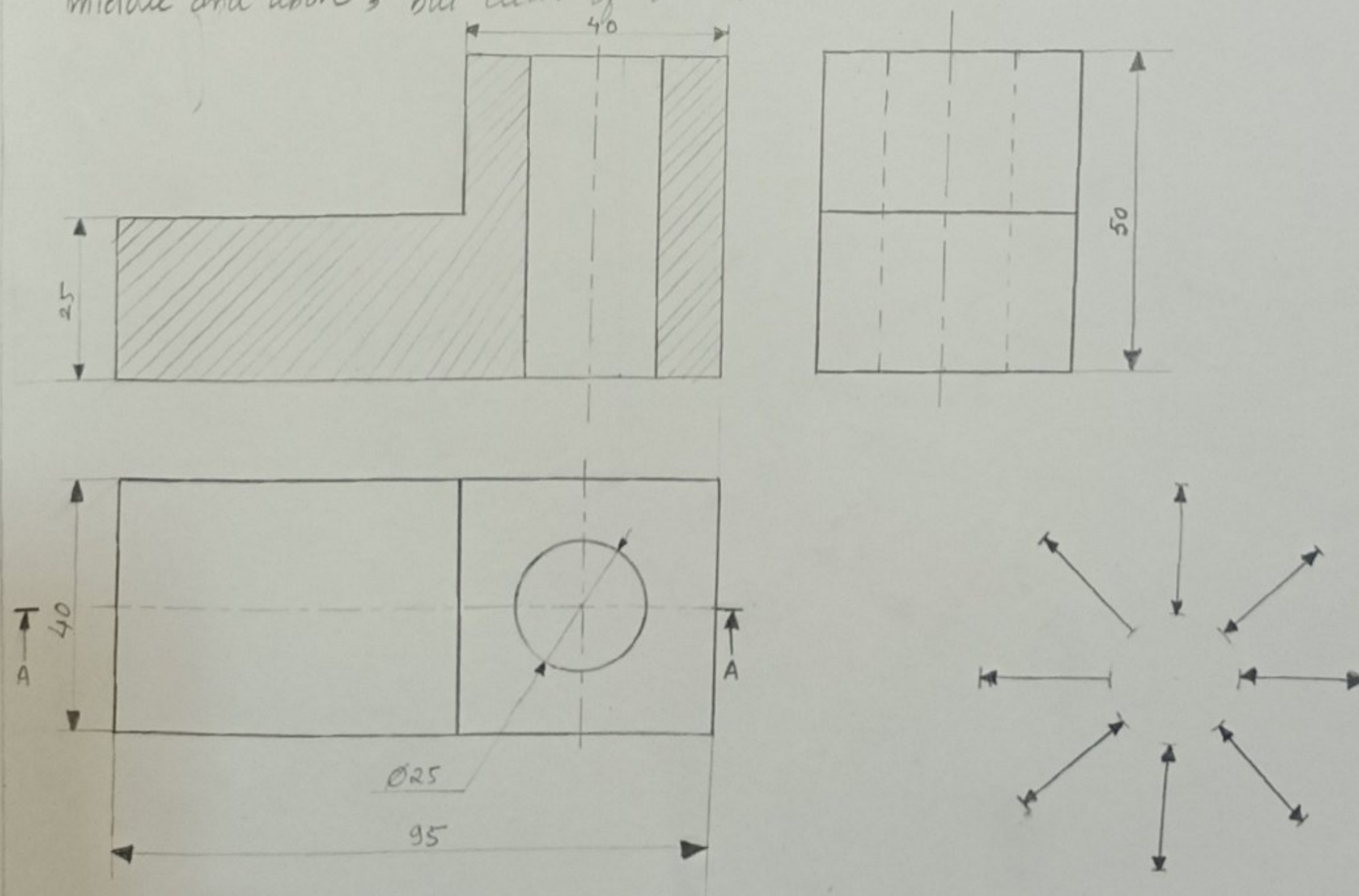


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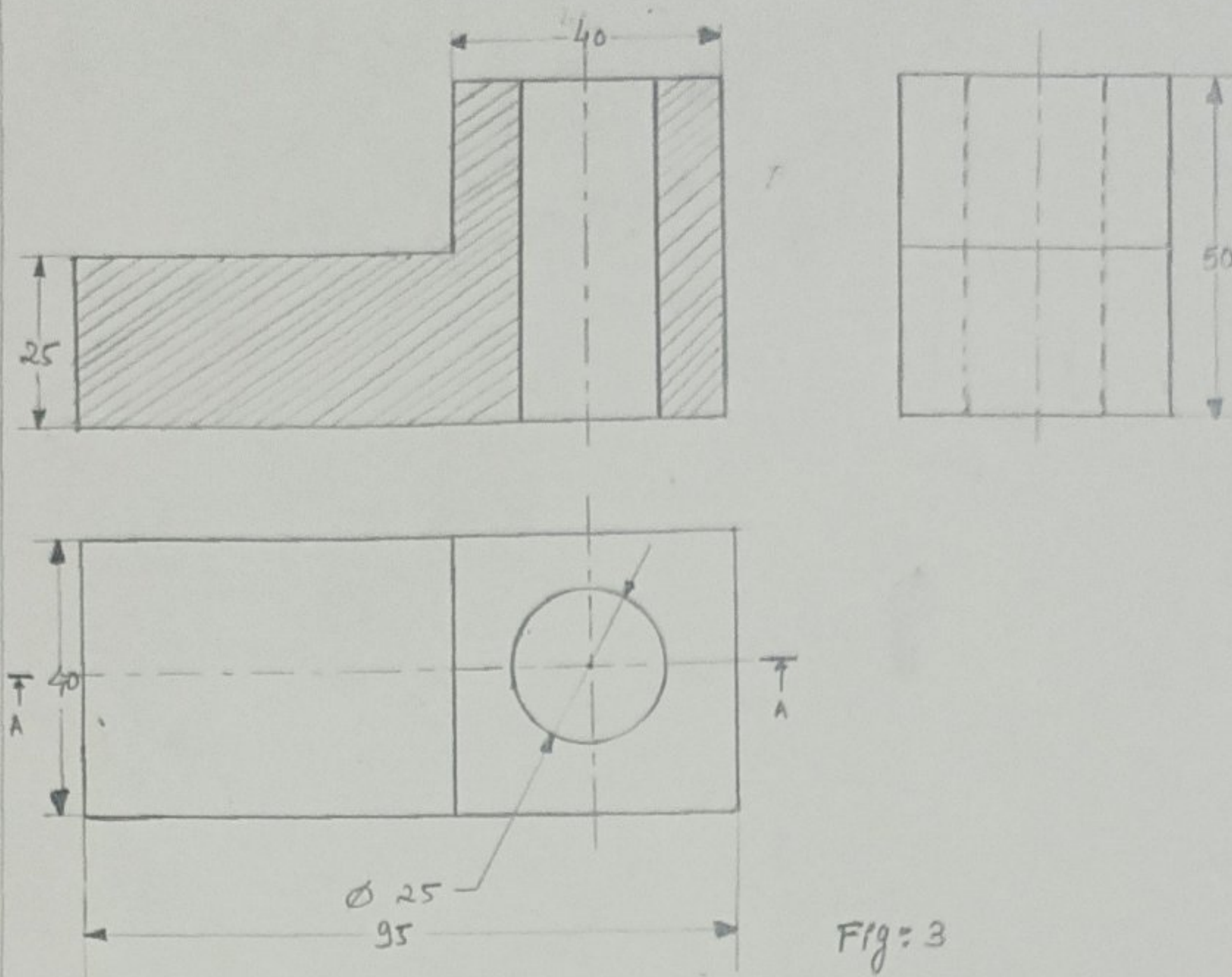


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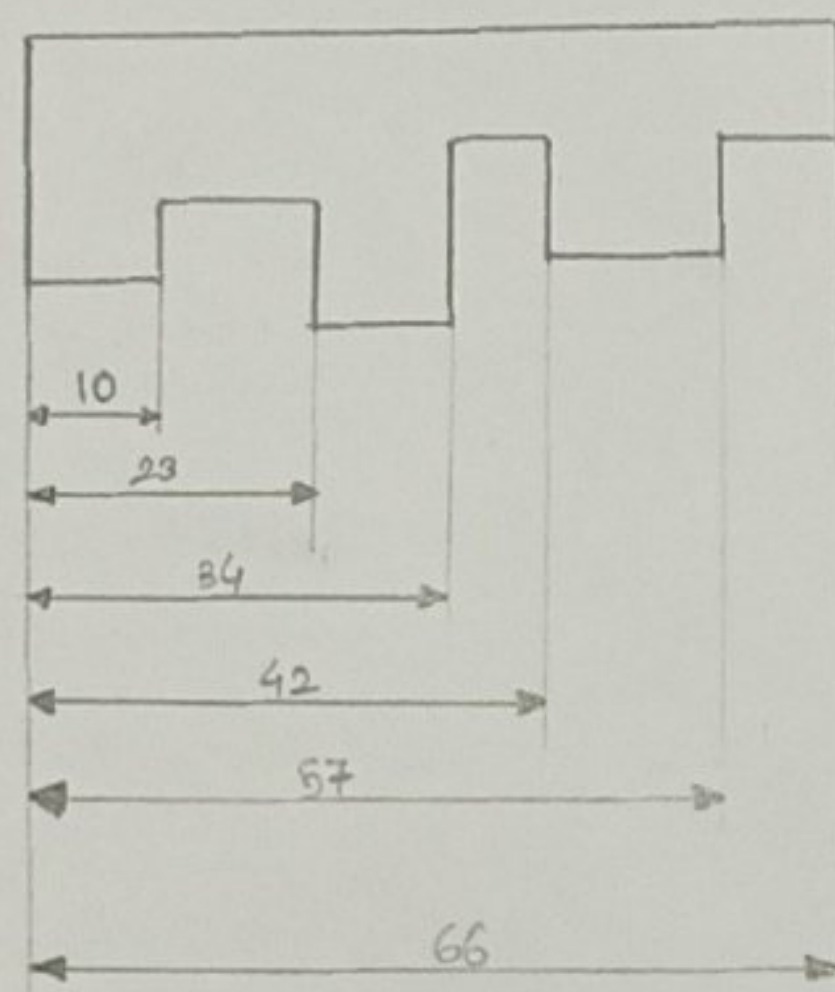


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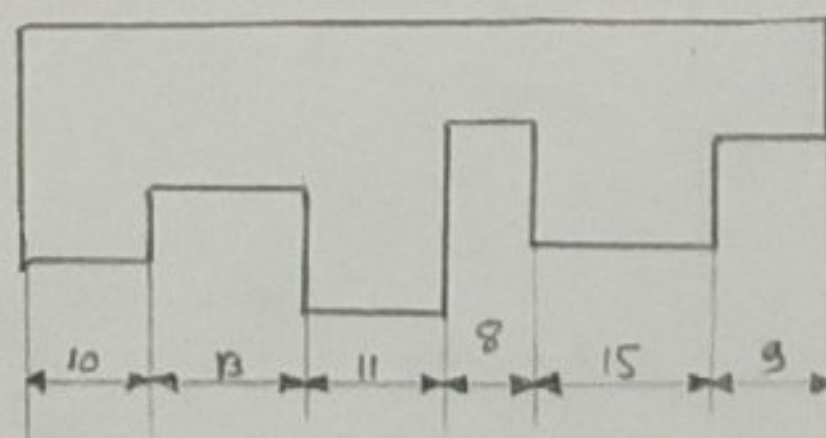


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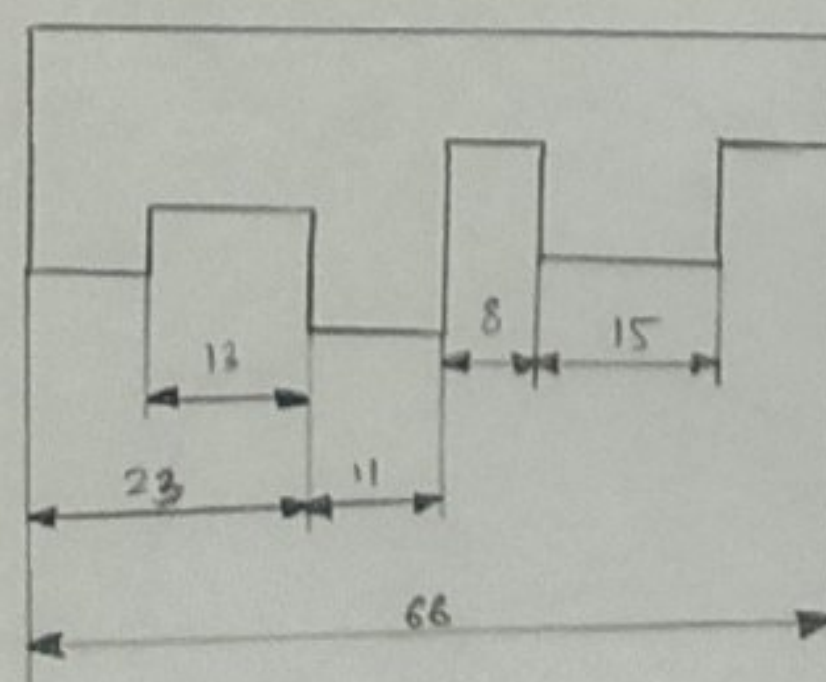


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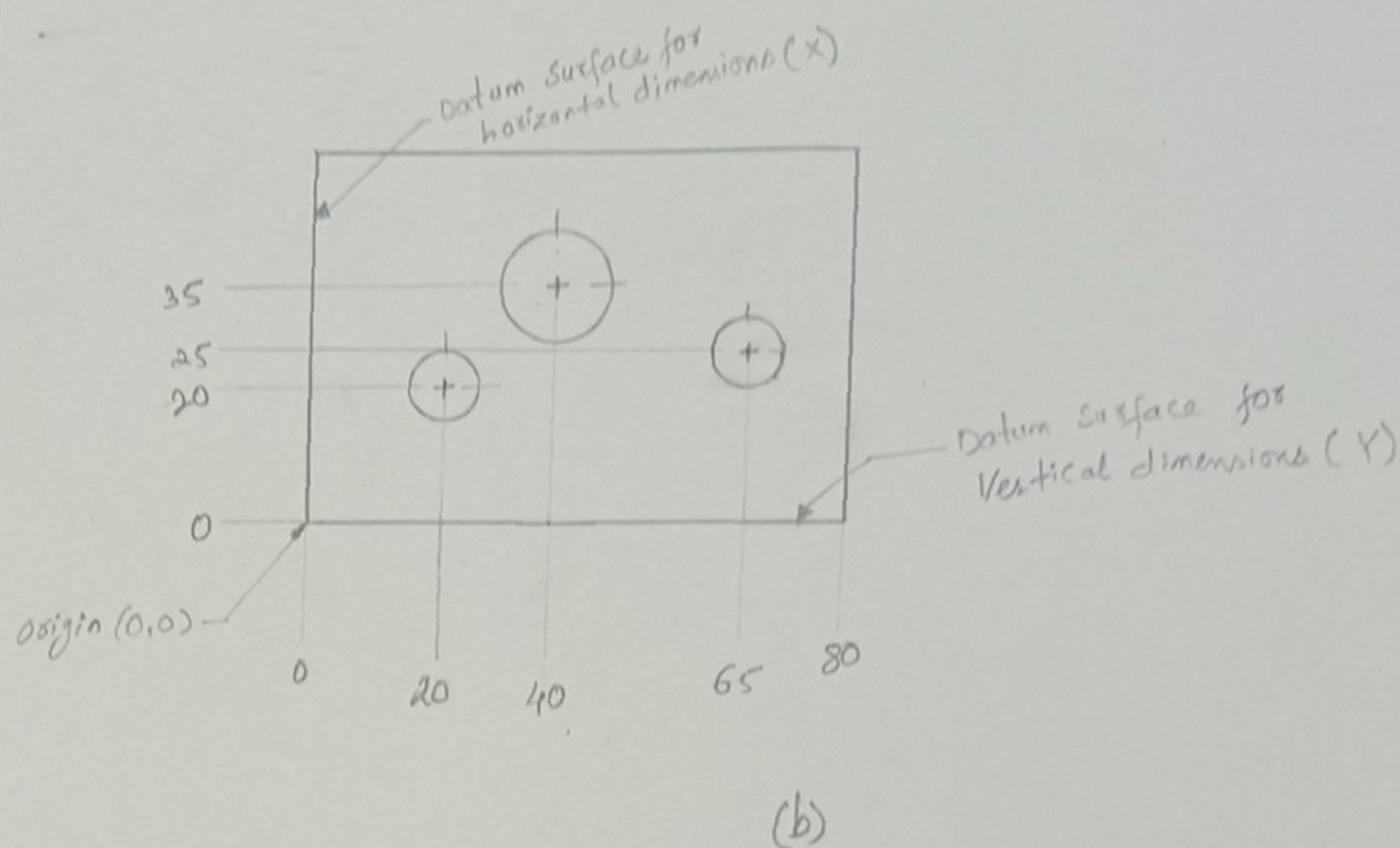
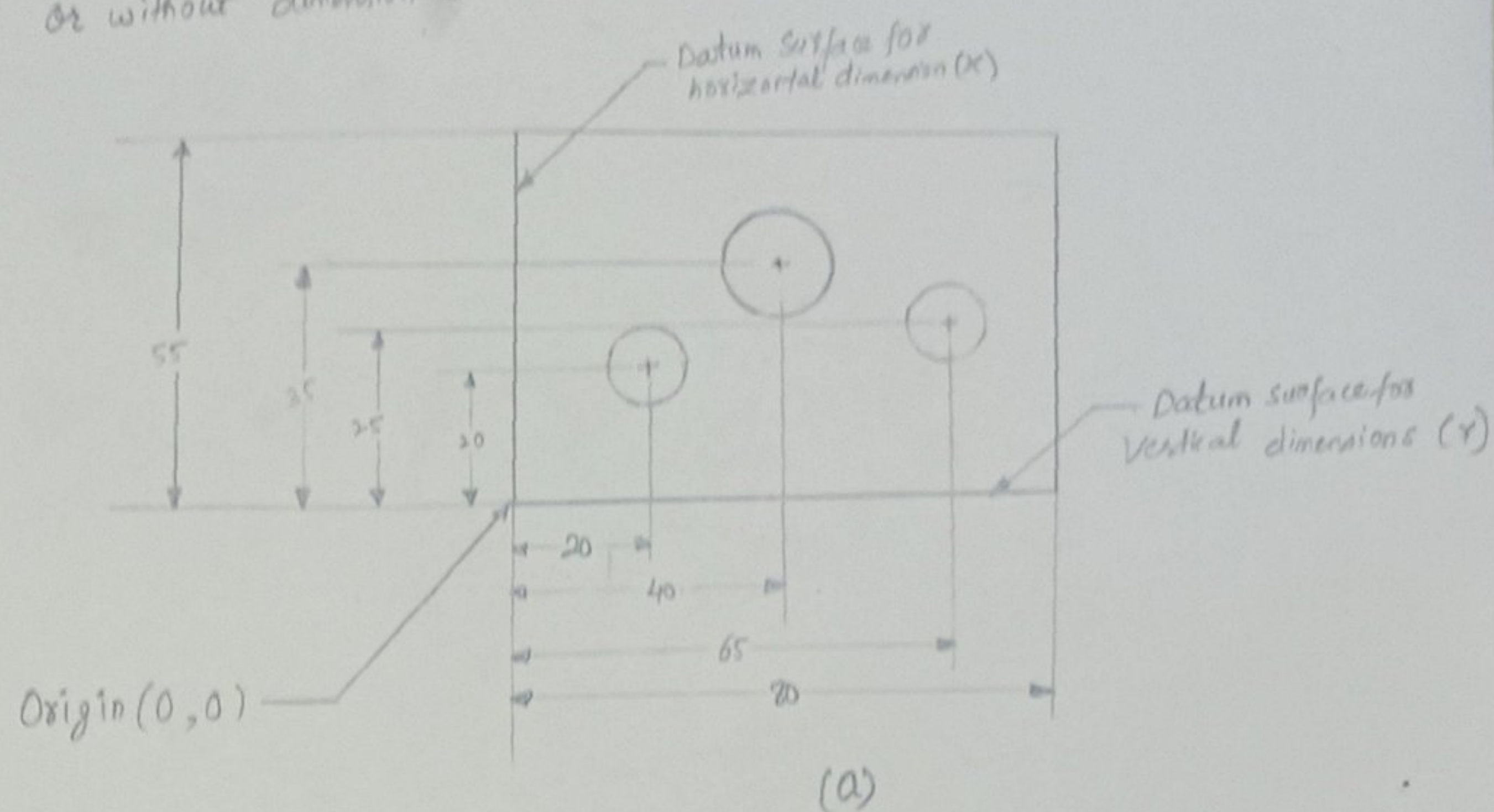


Fig = 7

GRAMIN TECH. & MANAG. CAMPUS VISNUPURI NED

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CLASS: FYBTECH (Comp. Engg) ROLL NO: 24COB67

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